

SCH3U SOLUTIONS UNIT

NOTE: Links in italics may not currently work. Will go "LIVE" as the unit progresses.
Please note that the outline starts as a duplicate of a previous semester to serve as an overview but is likely to change including assignments.

This is a live document. It will be adjusted as we go.

Assessments	Information
	Complete quizzes as you complete the learning related to each topic,
Quiz 1 topics: Solutions, Solubility Curves, explaining solubility	In Class Date: May 13 ~25 mins in class Correct AND hand in Quiz with answers will be posted on Classroom at the end of the quiz day.
Quiz 2 topics: Double Displacements, and Solution Concentrations. For 3U students only - Net Ionic equations as well	In Class Date: May 22 ~25 mins in class Correct AND hand in Quiz with answers will be posted on Classroom at the end of the quiz day.
	This is larger learning portfolio item (will take more time & worth larger value to portfolio mark)
UNIT TEST	DATE: June 5 . • Use the topics in below plan to create your own test topic list- SUGGESTED: Do a mindmap of the unit topics to solidify connections and understanding. • Multiple Choice strategies • 4C Solutions Test Review F23 blended course.doc.(updated for Fall 2023 semester) • 3U TEST REVIEW -> Questions you do NOT need to be able to do for the test - ASK TO HAVE THIS UPDATED NEAR END OF THE UNIT: #16 & 17, (updated for Spring 2025 semester) • SCH3U Exam Practice Quizzes & Tools - See slides for Solutions Unit. Please ADD any helpful resources you find. • Use any practice options provided with each learning topic (redo prior quizzes & activities) • Try to create your own review topics list before referring to this:

Learning Topics and Pacing Guide

[Solutions Textbook Answers](#) Part 1

[Solutions Textbook Answers](#) Part 2

Keeping Track! You can copy this [Table of Contents](#) to link all digital materials & track all topics. Optional but recommended.

KEEPING UP	LEARNING TOPIC COLLECTIONS	Complete Topics in collection <u>from Left to Right in the row</u> , then proceed to the next collection, unless otherwise instructed. Each topic should be completed in ONE class unless otherwise indicated. Boxes Filled in <u>Blue are Level 4</u> & can be skipped if you are behind the pace of the course (must complete Level 4 material if you are aiming for an above 85% in the course). It WILL appear on assessments.			
3U Gases Unit Goal: Complete Section 1 & 2 Apr 28 : 3U Intro to Gases Unit → ALL Start Solution unit Are you ready (for Solutions unit) p262-263 # 1-15 May 1 : Intro to Solutions May 2: Culminating work period May 5: Polarity and IMF in Solutions May 6: 3U Raft Activity & Share <i>All Pipette practice (maybe rainbow activity)</i> May 7: (Golden Bat) May 8: Last Culminating Work period May 9: Solubility Curve lab (3U) May 12: Solubility Curves (3U) ** QUIZ after completion of this row Reminder to add regularly to your learning portfolio AND use Retrieval practice routinely.	Solubility	TOPIC: SOLUTIONS UNIT INTRO SCH3U Terms & Translations → Prior to unit start, complete Are you Ready for solutions unit(p262-263) # 1-15 LEARN: → GOAL: Complete Engage Section of Solutions Intro HD (3U/4C) F23 Practice: → {Next time} Prep for ALL Labs in Unit: READ over labs in unit: DIAGRAM in flowchart style steps of lab with any written details as needed. HIGHLIGHT steps done by Partner A in one colour & Partner B in different colour. AS NEEDED for homework P 266-268 to define terms like solute/solvent, homogeneous, aqueous, electrolyte, acid base (3U text: P 266-268) → Lab Exercise 6.1.1 (p. 270) → p. 269 #1-8 → p. 271 #1 GASES UNIT: Complete the KMT theory Slide to review. Make notes on the topic for your gases learning Optional help/extensions: Watch 3D Printing using SOLUTIONS (10x faster than regular 3D printing)	TOPIC: IMF, Polarity & Solubility (2 days) LEARN: → GOAL Day 1 : Complete remaining Sections of Solutions Intro HD → Begin the Raft activity → Make sure you know these terms and definitions, using your textbook: - Solubility - Saturated vs Unsaturated vs Supersaturated - Miscible vs Immiscible - High solubility vs Low solubility vs Insoluble → {Optional} Make a lava lamp at home, add a video with explanation of chemistry involved, to portfolio as th level 4 extra! Day 2: 3U BW Stoichiometry Review question → Rafts → Pipetting Practice for Curve lab → Solubility Crossword Practice (3U only): Day 1: RAFT Solubility Activity (Maybe graded depending on semester) Day 2: Solubility Forces Crossword Solubility & Forces - Crossword Answers	TOPIC: Solubility Curves (2 Days) LEARN: → Day 1: Solubility Curve Lab (ammonium chloride) for next class. Solubility Curve Lab Instructions Slides; NOTE: factors affecting dissolving is next class so stop at that point Day 2: → Solubility Curves Learning Guide Practice: → Complete Lab questions → Complete questions on second page of Solubility Curves Note → Prep for ALL remaining Labs in Unit: READ over labs in unit: DIAGRAM in flowchart style steps of lab with any written details as needed. HIGHLIGHT steps done by Partner A in one colour & Partner B in different colour. → Remember to continue your Gases learning Optional help/extensions 2-Solubility Curves Practice Work {L} - This sheet ha additional practice questions	

Gases Unit Goal: Complete Section 3 May 13: Quiz 1 & Factors affecting Dissolving May 14: Double Displacement & Solubility Table May 15: Displacement lab May 16: Net Ionic Equations (3U only)/ 4C Time on Organic Chemistry	Dissolving & reacting	TOPIC: Factors affecting Dissolving LEARN: -> Continue the Solubility Curves HD learning for this topic Optional help/extensions: -> . Factors Affecting Dissolving (print after lesson and read) • Factors affecting dissolving Video {Optional}	TOPIC: Double Displacement & the Solubility Table LEARN: → Complete : Solubility Table and Double Displacements HD (3U) F23 -> Print this off if you like, can use it on tests or use one on periodic table (TEACHER NOTE: REPLACE WITH SCAN OF MODIFIED NEXT TIME) Solubility Table {note, just like the activity series, there are slight variations between different charts. Use this one OR the one on your periodic table) Practice: -> Predict products for Double Displacement Lab -> Complete Predicting products worksheet (*Except #3 on second page*), the Second page is on double displacements.-> COMPLETE: -> Quia Double Displacement & Solubility Quiz Solutions, explaining solubility, Solubility Curves Optional help/extensions: -> Video: Predicting Products of Double Replacement Reactions	TOPIC: Double Displacement lab LEARN: ->Double Displacement Lab & the solubility table. NOTE : Save Q#6 until you cover net ionic equations. Text reading from lab questions: Sequential qualitative analysis.pdf Practice: Optional help/extensions: -> Virtual Results for bottom two data rows. Play at least the beginning of the video to understand what is going into each well, then you can fast forward to the end where you can see all the results. Record the results for all the reactions of silver nitrate and copper sulphate (reactions for the cobalt compound aren't available virtually).	TOPIC: Net Ionic Equations LEARN: -> Net Ionic Equations (Activity in Groups of 3) Practice: -> Net Ionic Equations. Answers on second page -> Complete Net equations for any 3 of the reactions from the lab -> p. 322 #1,2 Optional help/extensions: -> Ionic and Net ionic Equations Slides. Use these if you miss the class activity -> OR EDPUZZLE: Double Displacements. Net Ionic Equations & the Solubility Table
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Gases Unit Goal: Day 3 Assignment Complete Section 4 May 20: Solution Concentration Day 1 (W/W, V/V, ppm etc) May 21: Molarity & Dilutions May 22 : Quiz 2; Practice Gases & solutions time May 23: Dilutions & Acids & bases, pH May 26: Water Treatment & Water Quality Activity assigned	SOLUTION CONCENTRATIONS ** QUIZ towards end of this Learning Collection **Test is approaching	TOPIC: Concentrations of solutions (g/L, ppm, W/W, W/V etc) [3U Only] LEARN: -> Concentration & Molarity Learning Guide F23 • Concentration Activity • Optional - Use your peers: Concentration of Solutions (HIP Website) to learn and practice this topic. As needed • Notes & further practice for the topic are also in the Measuring Solutions note Practice: • SEE GASES UNIT GOAL THE THE LEFT & COMPLETE IT • Complete: Concentration Calculations Student summary sheet {L} • Practice Questions - Answers on last pg • SCH molarity only worksheet • ASSIGNED: Solutions & Molarity Quia when ready • More as needed: ◦ Complete practice options on above website ◦ Chapter 6.3 (All Optional) ◦ Text p. 284 #1-8, p. 287 #11-17, p. 290 #19-22, p. 284 #8, p. 288 #14-17, p. 290 #1-10 Optional help/extensions: ->	TOPIC: Molarity & Ion Concentration & Dilutions LEARN: --> 3U: Complete the Concentration Learning Guide → 3U: Continue on to Dilutions Learning Guide F23 -> Watch Calculating Ion Concentration Video -> Part 1 of Measuring Solutions and Reactions in Solution Sheet , This is notes and questions combined {use digitally or print}. Practice: Complete all indicated for solution concentration (1) and Dilutions (2) in the Measuring Solutions Doc. -> Molarity text q's: p. 284 #8, p. 287 # 14-17, p. 290 # 19-22, p. 290 3, 7-10 -> Practice for Ions -> Reactions in solutions sections of the Measuring Solutions sheet above -> Highly recommended: Dilutions of Stock Solutions Worksheet & Ans {L} Optional help/extensions: -> Concentrations Slides → Tyler DeWitt solves Molarity problems. -> Dilution of Solutions Text -> Serial Dilution Video (3 mins) Dilutions text q's: p.302 #1,2,4a,5a, p.309 #9,11,17a, p.306 #6-8, 3-5 p.310 #12, 18a ◦	TOPIC: Acids, Bases & pH (3U only_ LEARN: -> choose from the following options to learn ->Acids and Bases note: Properties & theories -> Intro to Acids & Bases Slides (Includes videos on Arrhenius definition, weak vs strong AND how to calculate pH on your calculator. Optional paper practice activity on last two slides) → Read text pg 362 - 370 Practice: -> pH Calculations Question Sheet (Ref p368-371) -> ADD to your notes as needed: -> Ionization vs dissociation -> Define Strong and Weak acids and bases in terms of % ionization or dissociation -> how to calculate pH . pH Questions: Text: p. 367 #2-6, 375 #1-6, 403 #5-10, 409 #23 Link to these pages Practice only (not portfolio): Acids & Bases Quia Optional help/extensions: -> {for interest} REVIEW FROM BONDING: Reactions that Produce Acids and Bases ->Song I'm all about that base (a fun video that might help you remember!)	TOPIC: Water Quality , Treatment & Activity Learn Wastewater Treatment video Water Treatment Video Videos note-fill in Case of Walkerton - Rd p 298. Summarize & describe several ways it could have been avoided. Activity - Complete research prior to in class drawing day 3U Our Water Supply Assignment_Student.doc Water Supply Assignment Town Drawing.docx Water project assessment rubric
GASES QUEST SOON May 27: Solution stoichiometry (all)	Stoich in Solutions and Titration	TOPIC: Solution Stoichiometry LEARN: Solution Stoichiometry Learning	TOPIC: Titration Lab (2 Days) Half the class will work on the Virtual Titration & half will work on the Lab each day	TOPIC: Titration Math • Titration Problems - Answer Key	

May 28: Titration Learning & practical (all) May 29: Titration Learning & practical Day 2 (all) June 2: Titration calculations time (all) June 3: In class town Drawing Day June 4 : Review Time June 5:: Unit Test June 6: Discuss Science Electives in Dept for Grade 12 Review Time (gases); Partial Pressures Heads up Tues June 10 - In class Gases Quest Day June 9: In day Gases summary and Partial Pressures.		Guide F23 Prep for Titration: MUST watch Setting up a titration video Other aspects for interest {optional}: Making a standard solution video Making a standard solution: Beaker method Hmwk: - Measuring Solutions and Reactions in Solution Sheet Part B (except #8) - complete stoichiometry and titration questions on any prior handouts (several have them) - Solutions take home Quiz: Stoichiometry, ion concentration and dilution (may be a take home quiz) - Extra Practice (highly advised): Solution Stoichiometry Worksheet Optional help/extensions: Solution Stoichiometry (print)	LEARN: F23 Virtual Titration 3U/4C - This will cover the concepts of titration and the math -> LAB Titration of vinegar 2019-20 - Optional help/extensions: Video - Acid Base Titrations -- Part 1 - Chemistry - this covers collecting and using data during a titration experiment.	• Extra Practice (highly advised): Solution Stoichiometry Worksheet • This sheet has a few problems that start with a data table so it's highly recommended you try at least those. Titration Calculation Worksheet.pdf •	
TEST! See Table at the top for Review Material and Test Date.					
		For the teacher only! {This will take ~ 2 classes} Copy paste this template as needed: TOPIC: This lesson covers concepts in: ASSIGNED: Include in your portfolio. Level 4: Include a (not fancy) mindmap <u>NO assigned portfolio item.</u> Optional/extensions:	TOPIC: LEARN: -> -> Practice: -> Optional help/extensions: Reference slides for topic ASSIGNED: Include in your portfolio.		

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